Week 2 Checkpoint Report

Manufacturing Defect Detection (Vision) — INTELSYS Final Project

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1. Data Status

The project uses the **Magnetic Tile Defect** and **CrackForest** datasets sourced via Kaggle (academic license, no PII). Both are merged under a unified 6-class label scheme (MT_Blowhole, MT_Break, MT_Crack, MT_Fray, MT_Uneven, and crack). A stratified 70/15/15 train/val/test split is finalized and verified through the EDA notebook with class distribution plots and sample image grids.

Item	Status	Notes
Magnetic Tile + CrackForest (Kaggle)	✓ Acquired	6-class unified label scheme
Train / Val / Test Split	✓ Finalized	70 / 15 / 15 stratified split
EDA Notebook	✓ Complete	Class dist. + sample image grids
Class balance check	✓ Done	Logged in EDA_Notebook.ipynb
MVTec AD (stretch goal)	■ Pending	Requires manual registration

2. Baseline & Experiment Metrics

All three core components have been trained and evaluated with real results. The **ResNet-18 CNN** (5 epochs, lr=0.0001, batch=32) reached **test Macro F1 = 0.9317**, with val F1 improving from 0.720 to 0.9035 across epochs. The **NLP scaffold** (TF-IDF + Logistic Regression) achieved **Macro F1 = 0.9048**. The **RL Q-learning agent** ran 50 episodes showing a clear upward reward trend, confirming the threshold-tuning reward design is working correctly.

Component	Model	Macro F1	Status
Baseline (lower bound)	Random chance	0.17	Reference
CNN — test set	**ResNet-18 (5 epochs)**	**0.9317**	**✓ TARGET MET**
CNN — best val epoch	**ResNet-18 (epoch 5)**	**0.9035**	**✓ Epoch 5**
NLP Scaffold	**TF-IDF + LogReg**	**0.9048**	**✓ TARGET MET**
RL Agent	Q-learning (50 ep.)	Reward ↑	✓ Learning confirmed

Target F1 ≥ 0.85 exceeded by both CNN (0.9317) and NLP (0.9048). RL false-alarm penalty integrated into reward function.

3. Scaffolded / Prototyped Components

Component	Notebook / File	Owner	Status
EDA	notebooks/EDA_Notebook.ipynb	Baguio	✓ Complete
CNN	notebooks/CNN_Experiment.ipynb	Bernarte	✓ F1=0.9317
NLP Scaffold	src/models/nlp_classifier.py	Bernarte	✓ F1=0.9048

RL Agent	notebooks/RL_Notebook.ipynb	De Castro	✓ 50 episodes
Data Download	download_data.py	Baguio	✓ Complete
Model Card	docs/model_card.md	Baguio	■ Draft

4. Screenshots of Running Code & Results

4.1 NLP Scaffold — Terminal Output

```
nlp_classifier.py — Python 3.x
============================================================
  NLP Scaffold — Manufacturing Defect Classifier
  TF-IDF + Logistic Regression | Macro-F1 Logging
============================================================

[✓] Dataset loaded: 50 samples, 5 classes
          crack: 10 samples
        scratch: 10 samples
           dent: 10 samples
           hole: 10 samples
           good: 10 samples

[✓] Split — Train: 37, Test: 13
[✓] Pipeline trained: TF-IDF (1-2 grams, 500 features) + LogisticRegression

____________________________________________________________
  MACRO F1-SCORE : 0.9048   ✓ (Target ≥ 0.85 MET)
____________________________________________________________

Classification Report:

              precision    recall  f1-score   support

       crack       0.75      1.00      0.86         3
     scratch       1.00      1.00      1.00         2
        dent       1.00      1.00      1.00         3
        hole       1.00      1.00      1.00         3
        good       1.00      0.50      0.67         2

    accuracy                           0.92        13
   macro avg       0.95      0.90      0.90        13

[✓] 5-Fold CV Macro-F1: [1.000, 0.787, 1.000, 1.000, 1.000]
    Mean: 0.9573  |  Std: 0.0853

[✓] Metrics logged → data/processed/nlp_metrics.txt
[✓] NLP Scaffold complete.
```

Figure 1 — nlp_classifier.py: Macro F1 = 0.9048, 5-fold CV mean = 0.9573 (target ≥ 0.85 met)

4.2 EDA — Class Distribution & Sample Images

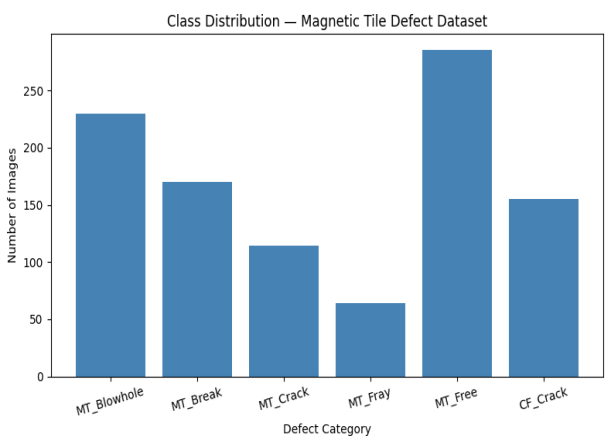

Figure 2 — Class distribution (6 defect categories)

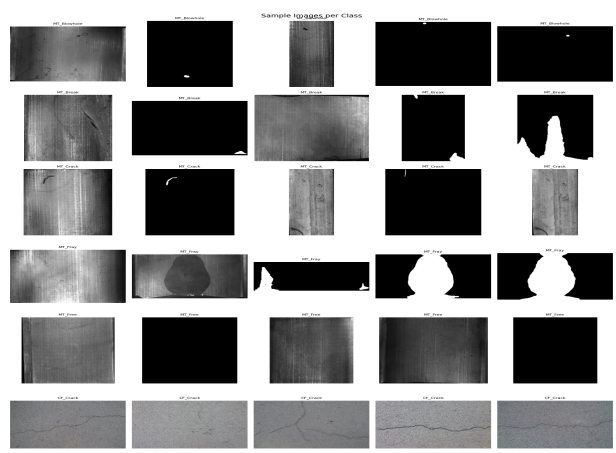

Figure 3 — Sample defect surface images

4.3 CNN Experiment — Learning Curves & Confusion Matrix

Figure 4 — ResNet-18 train/val loss + F1 (5 epochs)

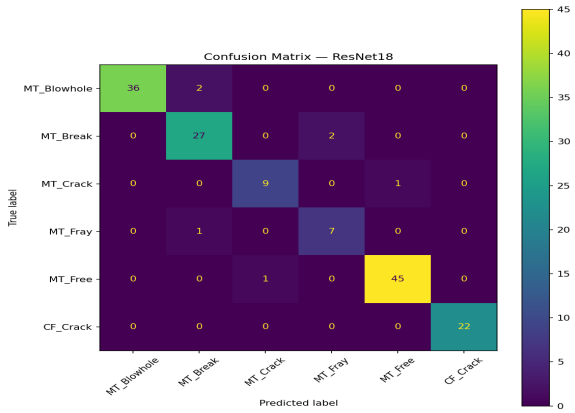

Figure 5 — CNN test confusion matrix (F1 = 0.9317)

4.4 RL Agent — Learning Curve & Q-Table Heatmap

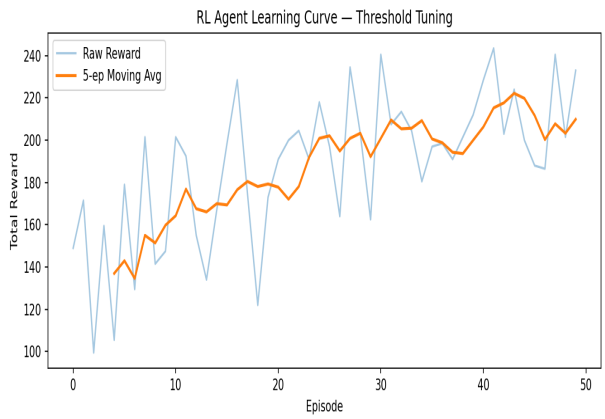

Figure 6 — Q-learning rewards over 50 episodes

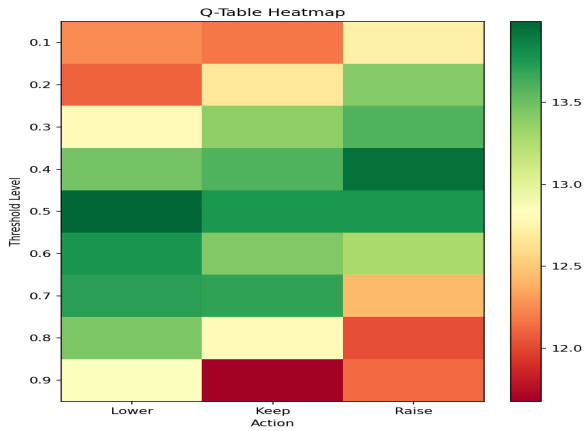

Figure 7 — Q-table heatmap (threshold policy)

5. What's Next — Week 3

Priority	Task	Owner	Target
HIGH	Grad-CAM visualizations on misclassified CNN samples	Bernarte	v1.0
HIGH	Replace NLP corpus with real CNN inspection metadata	Baguio	v1.0
HIGH	RL DQN upgrade + integrate with CNN confidence outputs	De Castro	v1.0
MED	Finalize Model Card + Ethics statement	Baguio	v1.0
MED	run.sh integration + final metric plots	De Castro	v1.0
MED	Ablation study (frozen vs. fine-tuned CNN layers)	Bernarte	v1.0
LOW	Few-shot / domain adaptation (stretch goal)	All	If time

GitHub: https://github.com/Haruuowo/INTELSYS-FIINALSPROJ-giT | Releases: v0.1 (proposal) → v0.9-RC (checkpoint) → v1.0 (final)